

Dopamine & Noradrenaline

Sérgio F genetic report
2 of October 2026

Name	Sérgio F
Sex	male (relevant to MAOA and MAOB, in chromosome X)
Report date	2 of October 2026 (local time)

What is: The same study of an example report "Dopamine & Norepinephrine Pathway" (example client is not Sérgio F), remade with DNA from Sérgio F: the same 25 genes and 48 SNPs, the same tags by allele (as in the original), the same themes, enhancers and recommendation logic.

Data: DNA raw chip (GRCh38) + local imputation (Beagle 5.5, full genome reference panel 30x, only genotypes with DR2 ≥ 0.8). Genetic data did not leave the local computer; only rsIDs and gene names were sent to public databases.

Important: are predispositions of isolated SNPs, not a diagnosis. The results are not classified as good or bad. All recommendations are suggestion (not applied) and should be decided with the doctor.

Contents

1. How it works and method	2
2. Dopamine pathway.....	3
3. Noradrenaline pathway	3
4. General view of the results and comparison with the example	5
5. Summary in 3 parts and reading as a system	8
6. Recommendations — suggestion (not applied)	11
7. Gene to gene, SNP to SNP	12
8. Drug purification.....	22
9. Laboratory markers suggested by the example	22
10. Limitations.....	23
11. Glossary.....	23

1. How it works and method

Dopamine and noradrenaline form a linked pathway: phenylalanine passes to tyrosine (PAH), L-DOPA (TH) and dopamine (DDC); dopamine then passes to noradrenaline (DBH). After release, they are recaptured by carriers (DAT/SLC6A3 and NET/SLC6A2), stored in vesicles (VMAT1/SLC18A1 and VMAT2/SLC18A2), act on receptors (DRD1-5; ADRA1A, ADRA2A, ADRB1-3) and are degraded by MAO-A, MAO-B and COMT, with the toxic aldehydes eliminated by ALDH2. The enzymes CYP2D6, CYP3A4 and CYP2C19 determine the clearance of drugs that act in this pathway.

- **Genes and SNPs:** exactly those of the example — via dopamine: 33 SNPs in 17 genes; via noradrenaline: 26 SNPs in 14 genes; 11 SNPs of 6 genes are in both (SLC18A1, COMT, MAOA, ALDH2, CYP2D6, CYP2C19). Total: 25 genes, 48 SNPs.
- **Sources:** 41 SNPs read directly on the chip; Impulsed 6 (Imputed DR2=...) 1 without data ("SNP not found in the data": rs28371706, CYP2D6*17).
- **Strand:** In all 48 SNPs the alleles of the example are in the strand forward (+), the same of the data — including the 24 SNPs of genes in the strand – (20 informative, all forward). **No genotype needed chain conversion.**
- **Palindromic SNPs (C/G or A/T):** rs1801253 (ADRB1), rs1042714 (ADRB2) and rs28386840 (SLC6A2) are in genes of strand +, without ambiguity; rs11564750 (SLC6A3) and rs1522305 (PAH) are in genes of strand – and were solved by the forward agreement of the example and by the frequency (C is the minority allele in forward, ≈ 10–13% in the reference population, as the example describes). None were left unsolved.
- **Chromosome X:** MAOA (2 SNPs) and MAOB (3 SNPs) appear with only one letter (man, hemizygous). There were no duplicate letters.
- **Heterozygous:** The labels of the two alleles (one copy of each) are shown. Genes with several SNPs: each SNP separately, without average.

Diagrams legend:  **Gene** Altered (some allele different from Typical/Normal)  **Gene** Typical  **Gene** Not determined  tags Increased ·  tags Reduced · ± both ·  only Typical/Normal  **AG** genotype (? = no data)  **Iron** enhancer indicated by the example. The example also uses the category "Better", which here is not used (without judgment good / bad).

2. Dopamine pathway

DOPAMINE PRODUCTION Phenylalanine → tyrosine → L-DOPA → dopamine (and phenylalanine → PEA)	○ PAH (GG) BH4 — Phenylalanine hydroxylase: converts phenylalanine into tyrosine, the first step for dopamine (use BH4). ↑ TH (CT) AG BH4 iron — Tyrosine hydroxylase: limiting enzyme that converts tyrosine into L-DOPA (using BH4 and iron). ↓ DDC (TT) CC CC — Decarboxylase of aromatic L-amino acids (AADC): converts L-DOPA into dopamine (and phenylalanine into PEA).
PEA PEA is degraded by MAO-B in PAA	↑ MAOB (G) G T — Monoamine oxidase B: degrades mainly dopamine and phenylethylamine (PEA) (gene in chromosome X).
LIBERTATION AND RECEPTORES Storage, release, recycling and receivers	↑ SLC6A3 (TT) GG CC — Dopamine transporter (DAT1): removes dopamine from synapse and recycles it to the neuron. ↓ SLC18A1 (GG) — VMAT1: carries monoamines (dopamine, norepinephrine, serotonin) to the storage vesicles. ↑ DRD1 (AA) CC TT — dopamine receptor D1, the most abundant in the brain. ○ DRD2 (GG) — Dopamine D2 receptor (rs1800497/Taq1A is in the neighboring ANKK1 gene, which regulates DRD2). ↑ DRD3 (TT) — Dopamine D3 receptor; involved in sleep-wake cycle. ± DRD4 (CT) — Dopamine D4 receptor. ↑ DRD5 (TT) — Dopamine receptor D5, with 10× higher affinity than D1.
DEGRADATION AND ELIMINATION Dopamine → DOPAL (MAO) → DOPAC (ALDH2) → HVA (COMT); or dopamine → 3-MT (COMT) → 3M, 4HPA (MAO-A) → HVA (ALDH2); HVA eliminated in urine	○ MAOA (T) T — Monoamine oxidase A: degrades dopamine, noradrenaline and serotonin (gene in chromosome X). ↑ COMT (GG) magnesium SAM-e — Catechol-O-methyltransferase: degrades dopamine, noradrenaline and adrenaline (uses SAM; magnesium). ○ ALDH2 (GG) Vitamins of complex B — aldehyde dehydrogenase 2: eliminates toxic aldehydes from degradation (DOPAL, DOPEGAL, MOHMA).
DRUG CLEARANCE Slow CYP keeps active drugs longer; Fast CYP eliminates them faster	○ CYP3A4 (GG) GG AA riboflavin iron — Cytochrome P450 3A4: metabolizes many psychiatric drugs. ○ CYP2C19 (GG) GG CC — Cytochrome P450 2C19: antidepressants and anxiolytics metabolism. ○ CYP2D6 (CC) GG ? riboflavin iron — Cytochrome P450 2D6: stimulants, antidepressants, antipsychotics and β-blockers.

In the dopamine pathway, MAOB appears twice in the example (PEA and dopamine degradation) — it is the same gene.

3. Noradrenaline pathway

NORADRENALINE PRODUCTION Dopamine → noradrenaline	○ DBH (CC) GG vitamin C copper — Dopamine β-hydroxylase: converts dopamine into noradrenaline (using vitamin C and copper).
LIBERTATION AND RECEPTORES Reuptake, storage and receivers	± SLC6A2 (CC) CT AA — Noradrenaline transporter (NET): recycles noradrenaline to the neuron. ↓ SLC18A1 (GG) — VMAT1: carries monoamines (dopamine, norepinephrine, serotonin) to the storage vesicles. ↓ SLC18A2 (CC) TT GG — VMAT2: carries noradrenaline (and other monoamines) to the vesicles. ± ADRA1A (AG) CT — Adrenergic receptor α1A. ○ ADRA2A (GG) — Adrenergic receptor α2A: stops the release of noradrenaline. ± ADRB1 (CC) AG β1 adrenergic receptor: frequency and contraction force of the heart. ↓ ADRB2 (GC) — β2 adrenergic receptor: airways, muscle and fat burning. ○ ADRB3 (AA) — β3 adrenergic receptor, especially in adipose tissue (fat and heat burning).
DEGRADATION AND ELIMINATION Noradrenaline → DOPEGAL (MAO-A) → DHPG (ALDH2) → MHPG (COMT); or → normethanephine (COMT) → MOHMA (MAO-A) → VMA (ALDH2); VMA eliminated in urine	○ MAOA (T) T — Monoamine oxidase A: degrades dopamine, noradrenaline and serotonin (gene in chromosome X). ↑ COMT (GG) magnesium SAM-e — Catechol-O-methyltransferase: degrades dopamine, noradrenaline and adrenaline (uses SAM; magnesium). ○ ALDH2 (GG) Vitamins of complex B — aldehyde dehydrogenase 2: eliminates toxic aldehydes from degradation (DOPAL, DOPEGAL, MOHMA).

DRUG CLEARANCE

Antidepressants, anxiolytics,
stimulants and β -blockers

○ **CYP2C19** GG GG CC — Cytochrome P450 2C19: antidepressants and anxiolytics metabolism.

○ **CYP2D6** CC GG ? riboflavin iron — Cytochrome P450 2D6: stimulants, antidepressants, antipsychotics and β -blockers.

4. General view of the results

Dopamine pathway — genes and SNPs

Gene	rsID	Genotype	Source	Result (example tag)	Example
DRD1	rs686	AA	chip	Increased DRD1 activity	GA
DRD1	rs5326	CC	chip	Typical DRD1 activity	CT
DRD1	rs4532	TT	chip	Increased DRD1 activity	CT
SLC18A1	rs1390938 Thr136Ile	GG	chip	Reduced SLC18A1 activity	GG
COMT	rs4680 Val158Met	GG	chip	Increased COMT activity	AG
DDC	rs921451	TT	chip	Typical DDC activity	CT
DDC	rs11575542	CC	chip	Typical DDC activity	CC
DDC	rs3735273	CC	chip	Reduced DDC activity	TC
SLC6A3	rs6347	TT	chip	Increased SLC6A3 activity	TT
SLC6A3	rs11564750	GG	chip	Typical SLC6A3 activity	GG
SLC6A3	rs27072	CC	chip	Increased SLC6A3 activity	CC
DRD2	rs1800497 Taq1A, ANKK1	GG	chip	Typical DRD2 activity	GA
MAOA	rs6323	T	chip	Typical MAOA activity (1 copy, X)	G
MAOA	rs909525	T	chip	Typical MAOA activity (1 copy, X)	C
ALDH2	rs671	GG	chip	Typical ALDH2 activity	GG
CYP2D6	rs3892097 CYP2D6*4	CC	chip	Typical CYP2D6 activity	CC
CYP2D6	rs1065852 CYP2D6*10	GG	chip	Typical CYP2D6 activity	GG
CYP2D6	rs28371706 CYP2D6*17	—	—	SNP not found in the data	GG
DRD4	rs1800955 -521C/T	CT	chip	1 Increased DRD4 activity + 1 allele Reduced DRD4 activity	TC
TH	rs6356	CT	chip	1 Allele Typical TH activity + 1 allele Increased TH activity	CC
TH	rs10770141	AG	chip	1 Increased TH activity + 1 Typical allele TH activity	AG
DRD5	rs6283	TT	imputation DR2=0.91	Increased DRD5 activity	CT
DRD3	rs6280 Ser9Gly	TT	chip	Increased DRD3 activity	TT
MAOB	rs3027452	G	chip	Increased MAOB activity (1 copy, X)	G
MAOB	rs2283729	G	chip	Increased MAOB activity (1 copy, X)	G
MAOB	rs1799836	T	chip	Increased MAOB activity (1 copy, X)	T
CYP3A4	rs35599367 CYP3A4*22	GG	chip	Typical CYP3A4 activity	GG
CYP3A4	rs12721627 CYP3A4*16	GG	chip	Typical CYP3A4 activity	GG
CYP3A4	rs55785340 CYP3A4*2	AA	chip	Typical CYP3A4 activity	AA
PAH	rs1522305	GG	chip	Typical PAH activity	GG
CYP2C19	rs4244285 CYP2C19*2	GG	chip	Typical CYP2C19 activity	GG
CYP2C19	rs4986893 CYP2C19*3	GG	chip	Typical CYP2C19 activity	GG

Gene	rsID	Genotype	Source	Result (example tag)	Example
CYP2C19	rs12248560 CYP2C19*17	CC	chip	Typical CYP2C19 activity	CC

Noradrenaline pathway — genes and SNPs

Gene	rsID	Genotype	Source	Result (example tag)	Example
SLC18A1	rs1390938 Thr136Ile	GG	chip	Reduced SLC18A1 activity	GG
COMT	rs4680 Val158Met	GG	chip	Increased COMT activity	AG
MAOA	rs6323	T	chip	Typical MAOA activity (1 copy, X)	G
MAOA	rs909525	T	chip	Typical MAOA activity (1 copy, X)	C
ALDH2	rs671	GG	chip	Typical ALDH2 activity	GG
CYP2D6	rs3892097 CYP2D6*4	CC	chip	Typical CYP2D6 activity	CC
CYP2D6	rs1065852 CYP2D6*10	GG	chip	Typical CYP2D6 activity	GG
CYP2D6	rs28371706 CYP2D6*17	—	—	SNP not found in the data	GG
CYP2C19	rs4244285 CYP2C19*2	GG	chip	Typical CYP2C19 activity	GG
CYP2C19	rs4986893 CYP2C19*3	GG	chip	Typical CYP2C19 activity	GG
CYP2C19	rs12248560 CYP2C19*17	CC	chip	Typical CYP2C19 activity	CC
ADRB1	rs1801253 Arg389Gly	CC	chip	Increased ADRB1 activity	CC
ADRB1	rs1801252 Ser49Gly	AG	chip	1 Increased ADRB1 activity + 1 allele Reduced ADRB1 activity	AA
ADRA1A	rs1048101	AG	chip	1 Allele Reduced ADRA1A activity + 1 allele Increased ADRA1A activity	GA
ADRA1A	rs1383914	CT	chip	1 Increased ADRA1A activity + 1 allele Reduced ADRA1A activity	CT
ADRB2	rs1042714 Gln27Glu, Q27E	GC	imputation DR2=1	1 Allele Reduced ADRB2 activity + 1 Normal allele ADRB2 activity	GC
ADRA2A	rs553668	GG	imputation DR2=1	Typical ADRA2A activity	GG
DBH	rs1611115 -1021C/T	CC	chip	Typical DBH activity	CC
DBH	rs1108580	GG	chip	Typical DBH activity	GG
SLC6A2	rs3785143	CC	chip	Typical SLC6A2 activity	CC
SLC6A2	rs3785157	CT	chip	1 Allele Reduced SLC6A2 activity + 1 allele Increased SLC6A2 activity	CC
SLC6A2	rs28386840 -3081A/T	AA	imputation DR2=0.98	Reduced SLC6A2 activity	TA
ADRB3	rs4994 Trp64Arg	AA	chip	Typical ADRB3 activity	AA
SLC18A2	rs363276	CC	imputation DR2=0.99	Typical SLC18A2 activity	CC
SLC18A2	rs363387	TT	imputation DR2=0.86	Typical SLC18A2 activity	TT
SLC18A2	rs363371	GG	chip	Reduced SLC18A2 activity	AG

Comparison with the example: The example has 48 SNPs, all genotyped in the example client. For Sérgio F there is a genotype in 47 of 48 (41 chip + 6 imputed); only the rs28371706 (CYP2D6*17) is missing. In 32 of the 47 SNPs, the genotype coincides with that of the example client (structural only comparison; "Example" column).

5. Summary in 3 parts

(1) Dopamine pathway — genes with different results from Typical/Normal

- **↑ DRD1** (AA) (CC) (TT) rs686 **AA** Enhanced DRD1 activity; rs5326 **CC**: Typical DRD1 activity; rs4532 **TT**: Increased DRD1 activity
- **↓ SLC18A1** (GG) rs1390938 (Thr136Ile) **GG**: Reduced SLC18A1 activity
- **↑ COMT** (GG) rs4680 (Val158Met) **GG**: Increased COMT activity
- **↓ DDC** (TT) (CC) (CC) rs921451 **TT** Typical DDC activity; rs11575542 **CC**: Typical DDC activity; rs3735273 **CC**: Reduced DDC activity
- **↑ SLC6A3** (TT) (GG) (CC) rs6347 **TT** Enhanced SLC6A3 activity; rs11564750 **GG**: Typical SLC6A3 activity; rs27072 **CC**: Increased SLC6A3 activity
- **± DRD4** (CT) rs1800955 (-521C/T) **CT**: 1 allele Enhanced DRD4 activity + 1 allele Reduced DRD4 activity
- **↑ TH** (CT) (AG) rs6356 **CT**: 1 allele Typical TH activity + 1 allele Increased TH activity; rs10770141 **AG**: 1 allele Increased TH activity + 1 allele Typical TH activity
- **↑ DRD5** (TT) rs6283 **TT**: Enhanced DRD5 activity
- **↑ DRD3** (TT) rs6280 (Ser9Gly) **TT**: Increased DRD3 activity
- **↑ MAOB** (G) (G) (T) rs3027452 G: Increased MAOB activity (1 copy, X); rs2283729 G: Increased MAOB activity (1 copy, X); rs1799836 T: Increased MAOB activity (1 copy, X)

(2) Noradrenaline pathway — genes with a different result from Typical/Normal

- **↓ SLC18A1** (GG) rs1390938 (Thr136Ile) **GG**: Reduced SLC18A1 activity
- **↑ COMT** (GG) rs4680 (Val158Met) **GG**: Increased COMT activity
- **± ADRB1** (CC) (AG) rs1801253 (Arg389Gly) **CC**: Increased ADRB1 activity; rs1801252 (Ser49Gly) **AG**: 1 allele Enhanced ADRB1 activity + 1 allele Reduced ADRB1 activity
- **± ADRA1A** (AG) (CT) rs1048101 **AG**: 1 allele Reduced ADRA1A activity + 1 allele Increased ADRA1A activity; rs1383914 **CT**: 1 allele Increased ADRA1A activity + 1 allele Reduced ADRA1A activity
- **↓ ADRB2** (GC) rs1042714 (Gln27Glu, Q27E) **GC**: 1 allele Reduced ADRB2 activity + 1 allele Normal ADRB2 activity
- **± SLC6A2** (CC) (CT) (AA) rs3785143 **CC** Typical SLC6A2 activity; rs3785157 **CT**: 1 allele Reduced SLC6A2 activity + 1 allele Enhanced SLC6A2 activity; rs28386840 (-3081A/T) **AA**: Reduced SLC6A2 activity
- **↓ SLC18A2** (CC) (TT) (GG) rs363276 **CC** Typical SLC18A2 activity; rs363387 **TT**: Typical SLC18A2 activity; rs363371 **GG**: Reduced SLC18A2 activity

SLC18A1 and COMT are in both routes and appear in both lists.

(3) Genes with Typical/Normal Result

- **○ DRD2** (GG) rs1800497 (Taq1A, ANKK1) **GG**: Typical DRD2 activity
- **○ MAOA** (T) (T) rs6323 T: Typical MAOA activity (1 copy, X); rs909525 T: Typical MAOA activity (1 copy, X)
- **○ ALDH2** (GG) rs671 **GG**: Typical ALDH2 activity
- **○ CYP2D6** (CC) (GG) (?) rs3892097 (CYP2D6*4) **CC**: Typical CYP2D6 activity; rs1065852 (CYP2D6*10) **GG**: Typical CYP2D6 activity; rs28371706 (CYP2D6*17) —: SNP not found in the data
- **○ CYP3A4** (GG) (GG) (AA) rs35599367 (CYP3A4*22) **GG**: Typical CYP3A4 activity; rs12721627 (CYP3A4*16) **GG**: Typical CYP3A4 activity; rs55785340 (CYP3A4*2) **AA**: Typical CYP3A4 activity
- **○ PAH** (GG) rs1522305 **GG**: Typical PAH activity
- **○ CYP2C19** (GG) (GG) (CC) rs4244285 (CYP2C19*2) **GG**: Typical CYP2C19 activity; rs4986893 (CYP2C19*3) **GG**: Typical CYP2C19 activity; rs12248560 (CYP2C19*17) **CC**: Typical CYP2C19 activity
- **○ ADRA2A** (GG) rs553668 **GG**: Typical ADRA2A activity
- **○ DBH** (CC) (GG) rs1611115 (-1021C/T) **CC**: Typical DBH activity; rs1108580 **GG**: Typical DBH activity
- **○ ADRB3** (AA) rs4994 (Trp64Arg) **AA**: Typical ADRB3 activity

SNP not found in the data (1 of 48): rs28371706 (CYP2D6*17). No gene was missing.

Genes with more than one SNP — coherence (each SNP shown separately, without average)

Gene	SNPs	Coherence
DRD1	rs686 AA: Enhanced DRD1 activity rs5326 CC: Typical DRD1 activity rs4532 TT: Enhanced DRD1 activity	without opposite meanings: Increased in 2 of 3 SNPs, Typical in the remaining — shown separately, without average
DDC	rs921451 TT: Typical DDC activity rs11575542 CC: Typical DDC activity rs3735273 CC: Reduced DDC activity	without opposite meanings: Reduced in 1 of 3 SNPs, Typical in the remaining — shown separately, without average
SLC6A3	rs6347 TT: Enhanced SLC6A3 activity rs11564750 GG: Typical SLC6A3 activity rs27072 CC: Enhanced SLC6A3 activity	without opposite meanings: Increased in 2 of 3 SNPs, Typical in the remaining — shown separately, without average
MAOA	rs6323 T: Typical MAOA activity (1 copy, X) rs909525 T: Typical MAOA activity (1 copy, X)	same sense (Typical/Normal)
CYP2D6	rs3892097 CC: Typical CYP2D6 activity rs1065852 GG: Typical CYP2D6 activity rs28371706 — SNP not found in the data	same sense (Typical/Normal)
TH	rs6356 CT: 1 allele Typical TH activity + 1 allele Increased TH activity rs10770141 AG: 1 allele Increased TH activity + 1 allele Typical TH activity	same meaning (Increased)
MAOB	rs3027452 G: Increased MAOB activity (1 copy, X) rs2283729 G: Increased MAOB activity (1 copy, X) rs1799836 T: Increased MAOB activity (1 copy, X)	same meaning (Increased)
CYP3A4	rs35599367 GG: Typical CYP3A4 activity rs12721627 GG: Typical CYP3A4 activity rs55785340 AA: Typical CYP3A4 activity	same sense (Typical/Normal)
CYP2C19	rs4244285 GG: Typical CYP2C19 activity rs4986893 GG: Typical CYP2C19 activity rs12248560 CC: Typical CYP2C19 activity	same sense (Typical/Normal)
ADRB1	rs1801253 CC: Enhanced ADRB1 activity rs1801252 AG: 1 allele Increased ADRB1 activity + 1 allele Reduced ADRB1 activity	non-concordant (there are Increased and Reduced alleles in different SNPs, without opposition between SNPs in 2 copies) — shown separately, without average
ADRA1A	rs1048101 AG: 1 allele Reduced ADRA1A activity + 1 allele Increased ADRA1A activity rs1383914 CT: 1 allele Increased ADRA1A activity + 1 allele Reduced ADRA1A activity	same pattern: each SNP with 1 allele Increased + 1 allele Reduced
DBH	rs1611115 CC: Typical DBH activity rs1108580 GG: Typical DBH activity	same sense (Typical/Normal)
SLC6A2	rs3785143 CC: Typical SLC6A2 activity rs3785157 CT: 1 allele Reduced SLC6A2 activity + 1 allele Increased SLC6A2 activity rs28386840 AA: Reduced SLC6A2 activity	non-concordant (there are Increased and Reduced alleles in different SNPs, without opposition between SNPs in 2 copies) — shown separately, without average
SLC18A2	rs363276 CC: Typical SLC18A2 activity rs363387 TT: Typical SLC18A2 activity rs363371 GG: Reduced SLC18A2 activity	without opposite meanings: Reduced in 1 of 3 SNPs, Typical in the remaining — shown separately, without average

There are no strict contradictory SNPs (two SNPs in 2 copies with opposite directions in the same gene). ADRB1 and SLC6A2 have Increased and Reduced alleles in different SNPs and therefore are shown side by side, without single conclusion.

Reading as a system (descriptive, without value judgment)

Dopamine production	PAH Typical; TH with an "Increased" copy in each of the 2 SNPs; DDC nonconcordant (rs921451 and rs11575542 Typical, rs3735273 Reduced). By the tags, it is more likely to produce close to the typical one, with a copy of higher activity in TH.
Storage	SLC18A1 (VMAT1) Reduced (GG); SLC18A2 (VMAT2) nonconcordant (2 SNPs Typical, rs363371 Reduced).
Reuptake	SLC6A3 (DAT) Increased in 2 SNPs and Typical in the third (same sense); SLC6A2 (NET) nonconcordant (rs28386840 Reduced, rs3785157 mixed, rs3785143 Typical).
Dopamine receptors	DRD1 Increased (2 SNPs) and Typical (1); DRD2 Typical; DRD3 Increased; DRD4 mixed; DRD5 Increased (imputed).

Degradation	COMT Increased (Val/Val, GG); MAOB Increased in 3 SNPs (1 copy, X); MAOA Typical (2 SNPs, X); ALDH2 Typical (elimination of aldehydes DOPAL/DOPEGAL).
Noradrenaline production	DBH Typical in the 2 SNPs.
Receptors of noradrenaline	ADRB1 Increased (rs1801253 CC) and mixed (rs1801252); ADRA1A mixed in 2 SNPs; Mixed ADRB2 (imputed); ADRA2A Typical (imputed); ADRB3 Typical.
Drug purification	CYP2D6 Typical (*4 and *10 absent); *17 not found in the data; CYP3A4 Typical (*22, *16 and *2 absent); CYP2C19 Typical (*2, *3 and *17 absent).

I read as a system and only by the labels of the example: the dopamine removal steps are with more "Increased" tags (reuptake by DAT, COMT Val/Val and MAO-B), while the production is close to the typical (PAH and DBH Typical, TH with an Increased copy) and the vesicular storage VMAT1 has "Reded" tag. Therefore, it is more likely to renew dopamine faster than the typical one, especially in the prefrontal cortex, where COMT weighs more. On the receiver side, D1, D3 and D5 have "Increased" tags and D2 is typical. In noradrenaline, production (DBH) and degradation (MAOA, ALDH2) are typical, but β 1 receptors have more "Increased" tags and NET non-concordant signals. This describes predispositions of isolated SNPs, each with a small effect: it is not a diagnosis nor says if something is better or worse.

6. Recommendations — suggestion (not applied)

The example orders the recommendations by probable impact (using analysis) and by evidence, and personalizes each one to concrete genes. Here we do not use analysis: a recommendation enters when an Sérgio F gene has the same tag that the example uses to customize it. Each line was crossed with the existing genetic plan.

#	Recommendation (example dose)	Sérgio F genes that activate it	Compatibility with the genetic plan	State
1	Protein in diet	ADRB2 : the protein can counteract this variant (sensitivity to leptin)	No conflict; quantity according to the plan/ doctor.	suggestion (not applied)
2	Green tea 400 mg (extract)	ADRB2 : moderate caffeine promotes adrenergic activity SLC6A3 : EGCG can increase extracellular dopamine (DAT and COMT)	Conflict with the genetic plan: green tea extract (EGCG) is left out (genetic liver PNPLA3/ MBOAT7/Pi*MZ probable); EGCG is catechol and adds to COMT Val/Val). If you want, only tea in infusion, counting caffeine.	suggestion (not applied)
3	Aerobic exercise 1 time	ADRB2 : improves sympathetic tone and physical form	No known conflict with the genetic plan.	suggestion (not applied)
4	Avoiding sugary foods and beverages	DRD3 : sugar can overactivate the D3 pathways	No known conflict with the genetic plan.	suggestion (not applied)
5	Curcumin 500 mg	MAOB : natural mild inhibitor of MAO-B SLC6A3 : slight inhibition of MAO and reuptake by DAT	Conflict with the genetic plan: pause curcumin is a suggestion (not applied; the real stack did not change) (UGT1A1*28/Gilbert, genetic liver and interaction with rapamycin via CYP3A4/P-gp). Do not reactivate because of this report.	suggestion (not applied)
6	Limit caffeine < 200 mg/dia	ADRB1 : excess increases sympathetic tone DRD3 : excess can over-stimulate the receptors and disturb the sleep	Coherent with the genetic plan: ≤ 200 mg/day, filtered coffee up to 12h (insomnia p80; CYP1A2 *1F A/C has normal elimination).	suggestion (not applied)
7	Magnesium 350 mg	ADRB1 : calms the sympathetic tone SLC18A1 : necessary for vesicular pumps (VMAT1)	No genetic conflict (TRPM6 without adjustment). Dose to decide with the doctor.	suggestion (not applied)
9	Rhodiola 500 mg	SLC6A3 : indirect effect on dopamine and reuptake	In vitro, rhodiola inhibits CYP3A4 (rapamycin) and CYP2C9 (*1/*3). Only with the doctor.	suggestion (not applied)
10	Stress management therapy 1 time/week	MAOB : regulates neurotransmitter renewal SLC6A3 : stress hormones can increase DAT activity	No known conflict with the genetic plan.	suggestion (not applied)
11	Avoiding pesticides	SLC18A1 : pesticides disturb VMAT1	No known conflict with the genetic plan.	suggestion (not applied)
12	Red fruits	MAOB : flavonoids with slight inhibition of MAO-B	Coherent with the genetic plan (polyphenols for food, not extracts).	suggestion (not applied)
14	Deep breathing 5 minutes	ADRB2 : can improve ADRB2 activity and physical form	No known conflict with the genetic plan.	suggestion (not applied)
17	Music therapy 30 minutes	MAOB : music temporarily rises dopamine and PEA	No known conflict with the genetic plan.	suggestion (not applied)
20	Regular sleep schedule	DRD3 : reduces overstimulation and improves energy	No known conflict with the genetic plan.	suggestion (not applied)
23	Yoga 30 minutes	ADRB1 : reduces the sympathetic tone	No known conflict with the genetic plan.	suggestion (not applied)
24	Zinc 15 mg	SLC6A3 : zinc can reduce DAT activity	Coherent with the genetic plan using the dose of the plan: total zinc 20–25 mg/day with copper 1.5–2 mg (not adding 15 mg extra).	suggestion (not applied)

Only by the enhancers (without genetic trigger): Diet iron — Conflict with the genetic plan: no iron supplements (HFE H63D, low ferritin target); Riboflavin (vitamin B2) — See genetic plan (S5: 5–10 mg or maintain 25 mg). (suggestion, not applied).

Recommendations of the example that do not activate in Sérgio F (the gene that personalizes them is Typical or in the opposite direction): Mindfulness (MAOA, DRD5); Cognitive training (DRD5); Bright light in the morning (MAOA); NAC (DRD2); Probiotics (DRD2); Resveratrol (MAOA).

Potentials and genetic plan: SAM-e (COMT potentiator) is not suggested because the genetic plan excludes it (COMT Val/Val + GNMT: plus SAM converted into homocysteine). Iron (TH, CYP): without iron supplements (HFE H63D). Vitamin C and copper (DBH): DBH is Typical; copper is already in the plan, along with zinc.

7. Gene to gene, SNP to SNP

↑ DRD1 Dopamine receptors

Dopamine receptor D1, the most abundant in the brain.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

1. rs686

Alleles (example)

A: Increased DRD1 activity

G: Reduced DRD1 activity

Genotype

AA

chip (DNA crude)

Increased DRD1 activity

Theme: Brain chemistry (D1 signaling); additions, mood and cognition (example: Brain chemistry)

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion.

Associations described in the example: Allele G (reduces DRD1 expression): more nicotine dependence, bipolar disorder, sleep bruxism and depressive symptoms; less opioid and alcohol dependence, schizophrenia and visual hallucinations in Parkinson's disease.

Coherence between DRD1 SNPs: without opposite meanings: Increased in 2 of 3 SNPs, Typical in the remaining — shown separately, without average

2. rs5326

Alleles (example)

C: Typical DRD1 activity

T: Reduced DRD1 activity

Genotype

CC

chip (DNA crude)

Typical DRD1 activity

Theme: Brain chemistry (D1 signaling); additions, cognition and planning (example: Brain chemistry)

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion.

Associations described in the example: Allele T (can lower dopamine in certain areas of the brain): addition to heroin, schizophrenia, POC and worse strategic planning.

Coherence between DRD1 SNPs: without opposite meanings: Increased in 2 of 3 SNPs, Typical in the remaining — shown separately, without average

3. rs4532

Alleles (example)

C: Reduced DRD1 activity

T: Enhanced DRD1 activity

Genotype

TT

chip (DNA crude)

Increased DRD1 activity

Theme: Brain chemistry (D1 signaling); attention (example: Brain chemistry)

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion.

Associations described in the example: Allele C (reduces DRD1 expression): treatment-resistant schizophrenia, more marked ADHD symptoms; also better accuracy in complex tasks and less autism.

Coherence between DRD1 SNPs: without opposite meanings: Increased in 2 of 3 SNPs, Typical in the remaining — shown separately, without average

↓ SLC18A1 Release and storage

VMAT1: carries monoamines (dopamine, norepinephrine, serotonin) to the storage vesicles.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

4. rs1390938 Thr136Ile

Alleles (example)

A: Increased SLC18A1 activity

G: Reduced SLC18A1 activity

Genotype

GG

chip (DNA crude)

Reduced SLC18A1 activity

Theme: levels of monoamines (example: monoamine levels)

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. In agreement with previous commercial reports.

Associations described in the example: Allele A (Thr136Ile plus monoamine transport by VMAT1): less autism, anxiety, affective and alcohol disorders, severe abstinence and bipolar disorder.

↑ COMT Degradation

Catechol-O-methyltransferase: degrades dopamine, noradrenaline and adrenaline (uses SAM; magnesium).

Potentials (example): magnesium, SAM-e · Blockers: not indicated by the example report

5. rs4680 Val158Met

Alleles (example)

A: Reduced COMT activity

G: Increased COMT activity

Genotype

GG

chip (DNA crude)

Increased COMT activity

Theme: resilience to stress (worrier/warrior); cognition and sensitivity to pain (example: "stress resilience")

Alleles of the example in the strand forward (+), such as data; without chain conversion. In agreement with previous commercial reports.

Associations described in the example: "Worrier/warrior": G (Val) gives COMT more active. GG ("warriors") degrade stress messengers faster, with better performance under stress and cognitive performance possibly lower at rest; AA ("worriers") the opposite; AG intermediate.

↓ DDC Dopamine production

Decarboxylase of aromatic L-amino acids (AADC): converts L-DOPA into dopamine (and phenylalanine into PEA).

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

6. rs921451

Alleles (example)

C: Reduced DDC activity

T: Typical DDC activity

Genotype

TT

chip (DNA crude)

Typical DDC activity

Theme: metabolism of L-DOPA (example: "L-DOPA metabolism")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. In agreement with previous commercial reports.

Associations described in the example: Allele C (less DDC): Parkinson's disease, worse motor response to L-DOPA, less orthostatic hypotension in Parkinson's disease and more nicotine dependence.

Coherence between DDC SNPs: without opposite meanings: Reduced in 1 of 3 SNPs, Typical in the remaining — shown separately, without average

7. rs11575542

Alleles (example)

C: Typical DDC activity

T: Reduced DDC activity

Genotype

CC

chip (DNA crude)

Typical DDC activity

Theme: metabolism of L-DOPA (example: "L-DOPA metabolism")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. In agreement with previous commercial reports.

Associations described in the example: T allele (less DDC): more marked somatic symptoms and higher alcohol consumption.

Coherence between DDC SNPs: without opposite meanings: Reduced in 1 of 3 SNPs, Typical in the remaining — shown separately, without average

8. rs3735273

Alleles (example)

C: Reduced DDC activity

T: Enhanced DDC activity

Genotype

CC

chip (DNA crude)

Reduced DDC activity

Theme: metabolism of L-DOPA (example: "L-DOPA metabolism")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. In agreement with previous commercial reports.

Associations described in the example: T allele (probably more DDC): better response to L-DOPA, more orthostatic hypotension in Parkinson's disease, more risk and severity of ADHD and less nicotine dependence. The C allele is the "Reduced".

Coherence between DDC SNPs: without opposite meanings: Reduced in 1 of 3 SNPs, Typical in the remaining — shown separately, without average

↑ SLC6A3 Dopamine release and receptors

Dopamine transporter (DAT1): removes dopamine from synapse and recycles it to the neuron.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

9. rs6347

Alleles (example)

C: Reduced SLC6A3 activity

T: Enhanced SLC6A3 activity

Genotype

TT

chip (DNA crude)

Increased SLC6A3 activity

Theme: dopamine reuptake; probability of ADHD (example: "odds of ADHD")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. In agreement with previous commercial reports.

Associations described in the example: T of rs6347, G of rs11564750 and C of rs27072: more risk of ADHD and more impulsivity and inattention; can increase the density of the dopamine transporter in certain regions.

Coherence between SLC6A3 SNPs: without opposite meanings: Increased in 2 of 3 SNPs, Typical in the remaining — shown separately, without average

10. rs11564750

Alleles (example)

G: Typical SLC6A3 activity

C: Reduced SLC6A3 activity

Genotype

GG

chip (DNA crude)

Typical SLC6A3 activity

Theme: dopamine reuptake; probability of ADHD (example: "odds of ADHD")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. SNP palindromic in a strand gene –: solved by the forward agreement of the example and by the minority frequency (C on forward). In agreement with previous commercial reports.

Associations described in the example: T of rs6347, G of rs11564750 and C of rs27072: more risk of ADHD; the example calls "Typical" to G and "Reduced" to C (rare allele).

Coherence between SLC6A3 SNPs: without opposite meanings: Increased in 2 of 3 SNPs, Typical in the remaining — shown separately, without average

11. rs27072

Alleles (example)

C: Increased SLC6A3 activity

T: Reduced SLC6A3 activity

Genotype

CC

chip (DNA crude)

Increased SLC6A3 activity

Theme: dopamine reuptake; probability of ADHD (example: "odds of ADHD")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. In agreement with previous commercial reports.

Associations described in the example: T of rs6347, G of rs11564750 and C of rs27072: more risk of ADHD and more impulsivity and inattention; can increase the density of the dopamine transporter.

Coherence between SLC6A3 SNPs: without opposite meanings: Increased in 2 of 3 SNPs, Typical in the remaining — shown separately, without average

○ **DRD2 Dopamine receptors**

Dopamine D2 receptor (rs1800497/Taq1A is in the neighboring ANKK1 gene, which regulates DRD2).

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

12. rs1800497 Taq1A, ANKK1

Alleles (example)

A: Reduced DRD2 activity

G: Typical DRD2 activity

Genotype

GG

chip (DNA crude)

Typical DRD2 activity

Theme: dopamine signaling (D2 receptors); reward and pleasure seeking (example: "dopamine signaling")

Alleles of the example in the strand forward (+), such as data; without chain conversion. In agreement with previous commercial reports.

Associations described in the example: Taq1A, in the ANKK1 gene, which controls DRD2. Allele A (A1): about 30% less D2 receptors in reward zones; associated with pleasure seeking (alcohol, food, substances), depression, chronic pain and PTSD, and also the most motivation.

○ **MAOA Degradation**

Monoamine oxidase A: degrades dopamine, norepinephrine and serotonin (gene in chromosome X).

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

13. rs6323

Alleles (example)

G: Increased MAOA activity

T: Typical MAOA activity

Genotype

T

chip (DNA crude)

Typical MAOA activity (hemizygous: 1 copy, chromosome X)

Theme: levels of monoamines (example: monoamine levels)

Alleles of the example in the strand forward (+), such as data; without chain conversion. Chromosome X: a single letter in the data, as expected in a man. In agreement with previous commercial reports.

Associations described in the example: Allele G: More active MAO-A and lower levels of dopamine, serotonin and noradrenaline. Variants of high and low activity have different psychiatric associations.

14. rs909525

Alleles (example)

C: Increased MAOA activity

T: Typical MAOA activity

Genotype

T

chip (DNA crude)

Typical MAOA activity (hemizygous: 1 copy, chromosome X)

Theme: levels of monoamines (example: monoamine levels)

Alleles of the example in the strand forward (+), such as data; without chain conversion. Chromosome X: a single letter in the data, as expected in a man. In agreement with previous commercial reports.

Associations described in the example: Allele C: MAO-A more active; it is often inherited in conjunction with rs6323 (may not be an independent factor).

○ **ALDH2 Degradation**

Aldehyde dehydrogenase 2: eliminates toxic aldehydes from degradation (DOPAL, DOPEGAL, MOHMA).

Potentials (example): Vitamins of complex B · Blockers: not indicated by the example report

15. rs671

Alleles (example)

A: Reduced ALDH2 activity

G: Typical ALDH2 activity

Genotype

GG

chip (DNA crude)

Typical ALDH2 activity

Theme: elimination of aldehydes (DOPAL/DOPEGAL) (example: "aldehyde clearance")

Alleles of the example in the strand forward (+), such as data; without chain conversion. In agreement with previous commercial reports.

Associations described in the example: Allele A: ALDH2 with reduced activity (50–70% with 1 copy, almost zero with 2), accumulation of acetaldehyde and aldehydes of dopamine/noradrenaline; very rare in the reference population.

○ CYP2D6 Drug purification

Cytochrome P450 2D6: stimulants, antidepressants, antipsychotics and β -blockers.

Potentials (example): riboflavin, iron · Blockers: not indicated by the example report

16. rs3892097 CYP2D6*4

Alleles (example)

C: Typical CYP2D6 activity

T: Reduced CYP2D6 activity

Genotype

CC

chip (DNA crude)

Typical CYP2D6 activity

Theme: drug clearance (example: "drug clearance")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion.

Associations described in the example: Allele T = CYP2D6*4, one of the most frequent non-functional variants.

17. rs1065852 CYP2D6*10

Alleles (example)

A: Reduced CYP2D6 activity

G: Typical CYP2D6 activity

Genotype

GG

chip (DNA crude)

Typical CYP2D6 activity

Theme: drug clearance (example: "drug clearance")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion.

Associations described in the example: Allele A = CYP2D6*10 (intermediate activity), rare in the reference population.

18. rs28371706 CYP2D6*17

Alleles (example)

A: Reduced CYP2D6 activity

G: Typical CYP2D6 activity

Genotype

—

SNP not found in
the data

SNP not found in the data

Theme: drug clearance (example: "drug clearance")

Searched in the chip by rsID, synonyms and position GRCh38 22:42129770, and in local imputation ($DR2 \geq 0.8$): absent.

Associations described in the example: Allele A = CYP2D6*17 (intermediate activity), rare in the reference population.

± DRD4 Dopamine receptors

Dopamine D4 receptor.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

19. rs1800955 -521C/T

Alleles (example)

C: Increased DRD4 activity

T: Reduced DRD4 activity

Genotype

CT

chip (DNA crude)

Mixed (heterozygous): 1 allele Increased DRD4 activity + 1 allele Reduced DRD4 activity

Theme: function of dopamine (search for novelty, extroversion) (example: "dopamine function")

Alleles of the example in the strand forward (+), such as data; without chain conversion.

Associations described in the example: Allele C (+40% of DRD4 expression): more search for novelty, extroversion and positive affect; less dependence on heroin and attention deficit; also more risk behavior and smoking.

↑ TH Dopamine production

Phenylalanine hydroxylase: limiting enzyme that converts tyrosine into L-DOPA (using BH4 and iron).

Potentials (example): BH4, iron · Blockers: not indicated by the example report

20. rs6356

Alleles (example)

C: Typical TH activity

T: Enhanced TH activity

Genotype

CT

chip (DNA crude)

Mixed (heterozygous): 1 allele Typical TH activity + 1 allele Increased TH activity

Theme: Cerebral chemistry (catecholamine production) (example: Brain chemistry)

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion.

Associations described in the example: T allele (may increase TH): late Parkinson's disease, schizophrenia, migraine, impulsivity in maltreated children.

21. rs10770141

Alleles (example)

A: Increased TH activity

G: Typical TH activity

Genotype

AG

chip (DNA crude)

Mixed (heterozygous): 1 allele Increased TH activity + 1 allele Typical TH activity

Theme: Cerebral chemistry (catecholamine production) (example: Brain chemistry)

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion.

Associations described in the example: Allele A (can increase TH expression): schizophrenia, less search for novelty, opioid dependence, more sensitivity to cocaine effects and greater cardiovascular reactivity to cold stress.

↑ DRD5 Dopamine receptors

Dopamine receptor D5, with affinity 10> higher than D1.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

22. rs6283

Alleles (example)

C: Reduced DRD5 activity

T: Enhanced DRD5 activity

Genotype

TT

imputation

DR2=0.91

Increased DRD5 activity

Theme: D5 signaling; probability of ADHD (example: "odds of ADHD")

Alleles of the example in the strand forward (+), such as data; without chain conversion. Absent from the chip; locally imputed (DR2=0.91; GP=0.0.02.0.98).

Associations described in the example: Allele C (minority): more risk of ADHD; may reduce the number or affinity of D5 receptors.

↑ DRD3 Dopamine receptors

D3 receptor of dopamine; involved in the sleep-wake cycle.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

23. rs6280 Ser9Gly

Alleles (example)

C: Reduced DRD3 activity

T: Enhanced DRD3 activity

Genotype

TT

chip (DNA crude)

Increased DRD3 activity

Theme: chronic fatigue; sleep-wake cycle (example: "odds of chronic fatigue")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. In agreement with previous commercial reports.

Associations described in the example: T allele (Ser9): associated with chronic fatigue syndrome; may increase the sensitivity or production of D3 receptor.

↑ MAOB Degradation

Monoamine oxidase B: degrades mainly dopamine and phenylethylamine (PEA) (gene in chromosome X).

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

24. rs3027452

Alleles (example)

A: Reduced MAOB activity

G: Increased MAOB activity

Genotype

G

chip (DNA crude)

Increased MAOB activity (hemizygous: 1 copy, chromosome X)

Theme: levels of monoamines; chronic fatigue (example: monoamine levels)

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. Chromosome X: a single letter in the data, as expected in a man. In agreement with previous commercial reports.

Associations described in the example: G rs3027452, G rs2283729 and T rs1799836: MAO-B more active, associated with chronic fatigue syndrome. The G of rs3027452 is also associated with the best response of mood to tryptophan.

25. rs2283729

Alleles (example)

A: Reduced MAOB activity

G: Increased MAOB activity

Genotype

G

chip (DNA crude)

Increased MAOB activity (hemizygous: 1 copy, chromosome X)

Theme: levels of monoamines; chronic fatigue (example: monoamine levels)

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. Chromosome X: a single letter in the data, as expected in a man. In agreement with previous commercial reports.

Associations described in the example: G rs3027452, G rs2283729 and T rs1799836: MAO-B more active, associated with chronic fatigue syndrome.

26. rs1799836

Alleles (example)

C: Reduced MAOB activity

T: Enhanced MAOB activity

Genotype

T

chip (DNA crude)

Increased MAOB activity (hemizygous: 1 copy, chromosome X)

Theme: levels of monoamines; chronic fatigue (example: monoamine levels)

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. Chromosome X: a single letter in the data, as expected in a man. In agreement with previous commercial reports.

Associations described in the example: G rs3027452, G rs2283729 and T rs1799836: MAO-B more active, associated with chronic fatigue syndrome; rs1799836 T is also associated with more anger.

CYP3A4 Drug purification

Cytochrome P450 3A4: metabolizes many psychiatric drugs.

Potentials (example): riboflavin, iron · Blockers: not indicated by the example report

27. rs35599367 CYP3A4*22

Alleles (example)

A: Reduced CYP3A4 activity

G: Typical CYP3A4 activity

Genotype

GG

chip (DNA crude)

Typical CYP3A4 activity

Theme: drug clearance (example: "drug clearance")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion.

Associations described in the example: Allele A = CYP3A4*22: about half of the activity, slower metabolism.

28. rs12721627 CYP3A4*16

Alleles (example)

A: Reduced CYP3A4 activity

G: Typical CYP3A4 activity

Genotype

GG

chip (DNA crude)

Typical CYP3A4 activity

Theme: drug clearance (example: "drug clearance")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion.

Associations described in the example: CYP3A4*16: the minority allele reduces activity. Note: the A/G tag, but in the text refers to the minority allele "C" (the *16 is C in the strand forward). G is "Typical" in both readings.

29. rs55785340 CYP3A4*2

Alleles (example)

A: Typical CYP3A4 activity

G: Reduced CYP3A4 activity

Genotype

AA

chip (DNA crude)

Typical CYP3A4 activity

Theme: drug clearance (example: "drug clearance")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion.

Associations described in the example: Allele G = CYP3A4*2: reduced activity.

PAH Dopamine production

Phenylalanine hydroxylase: converts phenylalanine into tyrosine, the first step for dopamine (use BH4).

Potentials (example): BH4 · Blockers: not indicated by the example report

30. rs1522305

Alleles (example)

G: Typical PAH activity

C: Increased PAH activity

Genotype

GG

chip (DNA crude)

Typical PAH activity

Theme: metabolism of phenylalanine; probability of schizophrenia (a study) (example: "odds of schizophrenia")

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. SNP palindromic in a strand gene –: solved by the forward agreement of the example and by the minority frequency (C on forward).

Associations described in the example: Allele C (minority): lower risk of schizophrenia in a study with more than 5000 people; the authors speculate that it increases PAH (without functional studies).

CYP2C19 Drug purification

Cytochrome P450 2C19: antidepressants and anxiolytics metabolism.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

31. rs4244285 CYP2C19*2

Alleles (example)

A: Reduced CYP2C19 activity

G: Typical CYP2C19 activity

Genotype

GG

chip (DNA crude)

Typical CYP2C19 activity

Theme: drug clearance (example: "drug clearance")

Alleles of the example in the strand forward (+), such as data; without chain conversion.

Associations described in the example: Allele A = CYP2C19*2: less ability to degrade drugs, including antidepressants.

32. rs4986893 CYP2C19*3

Alleles (example)

A: Reduced CYP2C19 activity

G: Typical CYP2C19 activity

Genotype

GG

chip (DNA crude)

Typical CYP2C19 activity

Theme: drug clearance (example: "drug clearance")

Alleles of the example in the strand forward (+), such as data; without chain conversion.

Associations described in the example: Allele A = CYP2C19*3: less ability to degrade drugs.

33. rs12248560 CYP2C19*17

Alleles (example)

C: Typical CYP2C19 activity

T: Enhanced CYP2C19 activity

Genotype

CC

chip (DNA crude)

Typical CYP2C19 activity

Theme: drug clearance (example: "drug clearance")

Alleles of the example in the strand forward (+), such as data; without chain conversion.

Associations described in the example: Allele T = CYP2C19*17: ultrafast metabolizer (lower effect of standard doses).**± ADRB1 Receptors of noradrenaline**Adrenergic receptor β 1: frequency and contraction force of the heart.**Potentials (example):** not indicated by the example report · Blockers: not indicated by the example report**34. rs1801253 Arg389Gly**

Alleles (example)

G: Reduced ADRB1 activity

C: Increased ADRB1 activity

Genotype

CC

chip (DNA crude)

Increased ADRB1 activityTheme: cardiovascular health; response to β -blockers (example: "cardiovascular health")

Alleles of the example in the strand forward (+), such as data; without chain conversion. SNP palindromic in a strand + gene: without ambiguity.

Associations described in the example: Allele C (Arg389): more activation of the β 1 receptor; associated with higher cardiovascular risk, higher LDL and lower exercise tolerance with training; also less adverse effects and better response to β -blockers.**Coherence between ADRB1 SNPs:** non-concordant (there are Increased and Reduced alleles in different SNPs, without opposition between SNPs in 2 copies) — shown separately, without average**35. rs1801252 Ser49Gly**

Alleles (example)

A: Increased ADRB1 activity

G: Reduced ADRB1 activity

Genotype

AG

chip (DNA crude)

Mixed (heterozygous): 1 allele Increased ADRB1 activity + 1 allele Reduced ADRB1 activityTheme: cardiovascular health; response to β -blockers (example: "cardiovascular health")

Alleles of the example in the strand forward (+), such as data; without chain conversion.

Associations described in the example: Allele A (Ser49): more stability of the β 1 receptor; associated with more cardiovascular risk; also better response to β -blockers.**Coherence between ADRB1 SNPs:** non-concordant (there are Increased and Reduced alleles in different SNPs, without opposition between SNPs in 2 copies) — shown separately, without average**± ADRA1A Receptors of noradrenaline**Adrenergic receptor α 1A.**Potentials (example):** not indicated by the example report · Blockers: not indicated by the example report**36. rs1048101**

Alleles (example)

A: Reduced ADRA1A activity

G: Increased ADRA1A activity

Genotype

AG

chip (DNA crude)

Mixed (heterozygous): 1 allele Reduced ADRA1A activity + 1 allele Increased ADRA1A activity

Theme: adrenaline signaling; probability of chronic fatigue (e.g.: "odds of chronic fatigue")

Alleles of the example in the strand forward (+), such as the data (gene in the strand —); without chain conversion.

Associations described in the example: Rs1048101 and T of rs1383914: associated with chronic fatigue (signalization of adrenaline more difficult); rs1048101 also the most incapacity in fibromyalgia.**37. rs1383914**

Alleles (example)

C: Increased ADRA1A activity

T: Reduced ADRA1A activity

Genotype

CT

chip (DNA crude)

Mixed (heterozygous): 1 allele Increased ADRA1A activity + 1 allele Reduced ADRA1A activity

Theme: adrenaline signaling; probability of chronic fatigue (e.g.: "odds of chronic fatigue")

Alleles of the example in the strand forward (+), such as the data (gene in the strand —); without chain conversion.

Associations described in the example: Rs1048101 and rs1383914: associated with chronic fatigue; rs1383914 also with fibromyalgia.

↓ ADRB2 Receptors of noradrenaline

Adrenergic receptor $\beta 2$: airways, muscle and fat burning.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

38. rs1042714 Gln27Glu, Q27E

Alleles (example)

G: Reduced ADRB2 activity

C: Normal ADRB2 activity

Genotype

GC

imputation

DR2=1

Mixed (heterozygous): 1 allele Reduced ADRB2 activity + 1 Normal allele ADRB2 activity

Theme: physical form (exercise capacity) and fat metabolism (example: "fitness")

Alleles of the example in the strand forward (+), such as data; without chain conversion. Palindromic SNP, but heterozygous: the orientation does not change the result. Absent from the chip; locally imputed (DR2=1; GP=0.1,0). In agreement with previous commercial reports.

Associations described in the example: Allele G (Glu27): lower performance in exercise and VO2 max., more fat mass and signs of resistance to leptin in small studies, more cardiovascular events in coronary patients and Graves' disease; better responds to caloric restriction, with risk of iyo effect.

○ ADRA2A Receptors of noradrenaline

Adrenergic receptor $\alpha 2A$: stops the release of noradrenaline.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

39. rs553668

Alleles (example)

A: Increased ADRA2A activity

G: Typical ADRA2A activity

Genotype

GG

imputation

DR2=1

Typical ADRA2A activity

Theme: metabolic health (fasting glucose) (e.g.: Metabolic health)

Alleles of the example in the strand forward (+), such as data; without chain conversion. Absent from the chip; locally imputed (DR2=1; GP=0.0,1).

Associations described in the example: Allele A (more ADRA2A receptors): higher fasting glucose and higher risk of type 2 diabetes.

○ DBH Noradrenaline production

Dopamine β -hydroxylase: converts dopamine into noradrenaline (using vitamin C and copper).

Potentials (example): vitamin C, copper · Blockers: not indicated by the example report

40. rs1611115 -1021C/T

Alleles (example)

C: Typical DBH activity

T: Reduced DBH activity

Genotype

CC

chip (DNA crude)

Typical DBH activity

Theme: levels of noradrenaline (example: "norepinephrine levels")

Alleles of the example in the strand forward (+), such as data; without chain conversion.

Associations described in the example: T allele (less than DBH: more dopamine and less noradrenaline): more neuroticism, search for novelty, impulsivity and ADHD of the adult; fewer migraines.

41. rs1108580

Alleles (example)

A: Reduced DBH activity

G: Typical DBH activity

Genotype

GG

chip (DNA crude)

Typical DBH activity

Theme: levels of noradrenaline (example: "norepinephrine levels")

Alleles of the example in the strand forward (+), such as data; without chain conversion.

Associations described in the example: Allele A (less DBH): ADHD and schizophrenia; less alcohol dependence.

± SLC6A2 Release and receptors of noradrenaline

Noradrenaline transporter (NET): recycles noradrenaline to the neuron.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report

42. rs3785143

Alleles (example)

C: Typical SLC6A2 activity

T: Enhanced SLC6A2 activity

Genotype

CC

chip (DNA crude)

Typical SLC6A2 activity

Theme: reuptake of noradrenaline; probability of ADHD; response to atomoxetine (example: "odds of ADHD")

Alleles of the example in the strand forward (+), such as data; without chain conversion. In agreement with previous commercial reports.

Associations described in the example: T of rs3785143, rs3785157 and rs28386840: risk of ADHD and inattention (may increase NET and decrease noradrenaline). The T of rs3785143 is associated with the worst response to atomoxetine.

Coherence between SLC6A2 SNPs: non-concordant (there are Increased and Reduced alleles in different SNPs, without opposition between SNPs in 2 copies) — shown separately, without average

43. rs3785157

Alleles (example)

C: Reduced SLC6A2 activity

T: Enhanced SLC6A2 activity

Genotype

CT

chip (DNA crude)

Mixed (heterozygous): 1 allele Reduced SLC6A2 activity + 1 allele Increased SLC6A2 activity

Theme: reuptake of noradrenaline; probability of ADHD (example: "odds of ADHD")

Alleles of the example in the strand forward (+), such as data; without chain conversion. In agreement with previous commercial reports.

Associations described in the example: T of rs3785143, rs3785157 and rs28386840: risk of ADHD and inattention**Coherence between SLC6A2 SNPs:** non-concordant (there are Increased and Reduced alleles in different SNPs, without opposition between SNPs in 2 copies) — shown separately, without average**44. rs28386840 -3081A/T**

Alleles (example)

A: Reduced SLC6A2 activity

T: Enhanced SLC6A2 activity

Genotype

AA

imputation

DR2=0.98

Reduced SLC6A2 activity

Theme: reuptake of noradrenaline; probability of ADHD; response to methylphenidate (example: "odds of ADHD")

Alleles of the example in the strand forward (+), such as data; without chain conversion. SNP palindromic in a strand + gene: without ambiguity. Absent from the chip; locally imputed (DR2=0.98; GP=0.0,1). In agreement with previous commercial reports.

Associations described in the example: T of rs3785143, rs3785157 and rs28386840: risk of ADHD and inattention; children with ADHD carriers responded better to methylphenidate.**Coherence between SLC6A2 SNPs:** non-concordant (there are Increased and Reduced alleles in different SNPs, without opposition between SNPs in 2 copies) — shown separately, without average**○ ADRB3 Receptors of noradrenaline**Adrenergic receptor β 3, especially in adipose tissue (fat burning and heat).**Potentials (example):** not indicated by the example report · Blockers: not indicated by the example report**45. rs4994 Trp64Arg**

Alleles (example)

A: Typical ADRB3 activity

G: Reduced ADRB3 activity

Genotype

AA

chip (DNA crude)

Typical ADRB3 activity

Theme: Metabolic health (fat metabolism) (example: Metabolic health)

Alleles of the example in the strand forward (+), such as the data (gene in the strand –); without chain conversion. In agreement with previous commercial reports.

Associations described in the example: Allele G (Arg64): β 3 receptor with less cAMP production; the metabolic effects are not consistent in the reference population.**↓ SLC18A2 Release and storage**

VMAT2: carries noradrenaline (and other monoamines) to the vesicles.

Potentials (example): not indicated by the example report · Blockers: not indicated by the example report**46. rs363276**

Alleles (example)

C: Typical SLC18A2 activity

T: Reduced SLC18A2 activity

Genotype

CC

imputation

DR2=0.99

Typical SLC18A2 activity

Theme: vesicular loading; probability of PTSD (example: "odds of PTSD")

Alleles of the example in the strand forward (+), such as data; without chain conversion. Absent from the chip; locally imputed (DR2=0.99; GP=0.0,1).

Associations described in the example: T allele: more risk of PTSD.**Coherence between SLC18A2 SNPs:** without opposite meanings: Reduced in 1 of 3 SNPs, Typical in the remaining — shown separately, without average**47. rs363387**

Alleles (example)

G: Increased SLC18A2 activity

T: Typical SLC18A2 activity

Genotype

TT

imputation

DR2=0.86

Typical SLC18A2 activity

Theme: vesicular loading; probability of additions; inhibitory control (example: "odds of addictions")

Alleles of the example in the strand forward (+), such as data; without chain conversion. Absent from the chip; locally imputed (DR2=0.86; GP=1.0,0).

Associations described in the example: Allele G: less alcohol and nicotine dependence and better inhibitory control.**Coherence between SLC18A2 SNPs:** without opposite meanings: Reduced in 1 of 3 SNPs, Typical in the remaining — shown separately, without average

48. rs363371

Alleles (example)

A: Increased SLC18A2 activity

G: Reduced SLC18A2 activity

Genotype

GG

chip (DNA crude)

Reduced SLC18A2 activity

Theme: vesicular load; mental health (Parkinson, schizophrenia, ELA) (example: "mental health")

Alleles of the example in the strand forward (+), such as data; without chain conversion. In agreement with previous commercial reports.

Associations described in the example: Allele G: more Parkinson's disease and schizophrenia, minus ELA.

Coherence between SLC18A2 SNPs: without opposite meanings: Reduced in 1 of 3 SNPs, Typical in the remaining — shown separately, without average

8. Drug purification

Enzyme	SNPs of example	Interpretation
CYP2D6	*4 rs3892097 CC; *10 rs1065852 GG; *17 rs28371706 not found	Typical in available 2 SNPs. It processes stimulants, antidepressants, antipsychotics and β -blockers. The chip does not see deletions/duplications of the gene (CNV): the previous pharmacogenomic analysis gives probable normal metabolizer (*1/*2), to be confirmed with clinical testing before narrow margin drugs.
CYP3A4	*22 rs35599367 GG; *16 rs12721627 GG; *2 rs55785340 AA	Typical. It processes many psychiatric medications. (Outside this report: CYP3A5 is expressed, which matters to rapamycin and tacrolimus — see pharmacogenomic analysis.)
CYP2C19	*2 rs4244285 GG; *3 rs4986893 GG; *17 rs12248560 CC	Typical (normal metabolizer, *1/*1). It processes antidepressants and anxiolytics.

9. Laboratory markers suggested by the example

The example includes an analysis section to verify. This report is only genetic and did not use analyzes or values; there are only names, as a reference to discuss with the doctor:

Fasting Glucose, Iron, Cortisol (saliva), Homovanilic Acid (HVA, urine), Noradrenaline (plasma and urine), Prolactin, Vanillmandelic Acid (VMA, urine), Vitamin B12, Vitamin B2, Vitamin D (25-OH), Zinc, Adiponectin, Leptin, Heart Rate at rest, Magnesium, VO2 max., Copper, Folate, HbA1c, HOMA-IR, Homocysteine, Insulin in fasting, SAM-e, Total Glutathione, Vitamin B6, Vitamin C.

10. Limitations

- They are predispositions of isolated SNPs, with small effects and often small studies or a single population. They do not allow clinical conclusions by themselves and are not a diagnosis.
- The labels, themes and associations are those of the example report, not an independent literature review.
- The example has internal inconsistencies: in the diagram, DRD5 appears as "Higher Activity" when the page of SNP says "reduced DRD5 activity"; SLC6A2 and SLC18A2 appear as "Typical" in the diagram despite non-typical SNPs; in rs12721627 (CYP3A4*16) the tags are A/G but the text speaks of the allele "C". Here is always followed by allele of the page of each SNP.
- Chip: CYP2D6 has no deletions/duplications (CNV, *5, *xN) or most of the rare alleles; rs28371706 (*17) is not in the data. The CYP2D6 status does not replace a clinical pharmacogenetic test.
- Imputation: Imputational 6 SNPs (DR2 between 0.86 and 1.00: DRD5 rs6283, ADRB2 rs1042714, ADRA2A rs553668, SLC6A2 rs28386840, SLC18A2 rs363276 and rs363387). The estimated accuracy is about 95% in common variants, but an imputed genotype may be wrong.
- DRD2 rs1800497 (Taq1A) is in the ANKK1 gene; the binding to D2 receptors is indirect.
- Haplotypes not evaluated: some SNPs inherit together (e.g. rs6323/rs909525 of MAOA; rs686/rs4532 of DRD1), so they should not be added as independent signals.
- Out of the reach of the chip: repetitions VNTR (DAT1 40 pb, DRD4 48 pb, MAOA uVNTR), rare variants and structural changes.
- The doses of the example are generic. This report does not use analysis; any change in the stack should be decided with the doctor.

11. Glossary

3-Metoxitiramina (3-MT) — product of dopamine degradation by COMT.

3M,4HPA — intermediate aldehyde metabolite of 3-MT, then eliminated by ALDH2.

ADHD — hyperactivity disorder and attention deficit.

Hydrogen — reactive by-product of dopamine and noradrenaline degradation; some are toxic if not eliminated.

β-blockers — drugs that block noradrenaline/adrenaline receptors, lowering heart rate.

DHPG — safer metabolite, formed when ALDH2 eliminates DOPEGAL.

DOPAC — stable product of dopamine degradation, after eliminating DOPAL.

DOPAL — toxic aldehyde formed by MAO from dopamine; eliminated by ALDH2.

DOPEGAL — toxic aldehyde formed by MAO from noradrenaline; eliminated by ALDH2.

HVA (homovanil acid) — the main final product of dopamine, excreted in urine.

L-DOPA — direct precursor of dopamine, made from tyrosine.

MHPG — main product of noradrenaline degradation.

MOHMA — intermediate aldehyde of normetanephrine; ALDH2 converts it into VMA.

Normethanephrine — metabolite of noradrenaline formed by COMT.

PAA — phenylacetic acid, a product of PEA degradation by MAO-B.

PEA (phenylethylamine) — stimulant compound made from phenylalanine; rapidly degraded by MAO-B.

VMA (vanillylmandelic acid) — the main final product of noradrenaline, excreted in urine.

Vesicle — small pouch inside the neuron that stores neurotransmitters until release.

Hemizygous — with a single copy of the gene (X chromosome genes in a man).

DR2 — Imputation confidence measure (0 to 1); here only genotypes with DR2 ≥ 0.8 are used.